

CHLOROPHYLL

Chlorophyll is a vital molecule because it captures sunlight. It is not the only pigment found in plants, but it is the most common. The other pigments absorb different colors of light than chlorophyll but can still be used for photosynthesis.

Changing colors

In the fall, the days shorten and there is less light for photosynthesis. Some trees and shrubs prepare for this by losing their leaves. The chlorophyll starts to break down, revealing the yellow and orange pigments that are also in the leaf. Any sugar that is trapped in the leaf is used to make red, purple, and crimson pigments.



In the fall, leaves change colors.

This is how plants make energy:

Carbon dioxide + water + light = glucose + oxygen

This process is called

PHOTOSYNTHESIS.



MELVIN
CALVIN

American scientist **Melvin Calvin** discovered how the dark reaction in photosynthesis works—in 1961 he won the Nobel prize for chemistry.

LIGHT...

Photosynthesis is carried out in two stages: the light and dark reactions. In the light reaction, the sunlight is captured by the chlorophyll and its energy is used to make ATP, a molecule that transports energy around the cell (see pp.16–17). During this stage water molecules are split to produce oxygen, which is then released back into the atmosphere through the stomata under the leaves.

... AND DARK

Away from the light, the ATP is used to provide energy in the dark reaction (also known as the Calvin cycle), which converts carbon dioxide into glucose. Some glucose is used by the plant cells to carry out their normal functions. The rest is converted into a more complex sugar called starch, which can be stored. Starch can be turned back into glucose when the plant needs it.

in every square millimeter of leaf.